

• • LIVE MASTERCLASS

Calculus in 3 Hours

Derivatives and integrals — actually understood. Geometric intuition first, formulas second. You'll leave knowing why the rules work, not just what they are.

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹399	3 Hours	Intermediate	Live	1st Saturday, 11 AM



Prasant Mishra

Founder & Instructor, Qurious Academy

What is Calculus in 3 Hours?

Most people who were taught calculus in school or college learned the rules — the power rule, the chain rule, integration by substitution — without ever understanding what they're really doing geometrically. This session reverses that. We start with the visual intuition: what a derivative really is (the slope of a curve at a point), what an integral really is (accumulated area), and why the Fundamental Theorem connects them. The formulas follow naturally once the picture is clear.

This is a focused three-hour session covering the two big ideas in calculus — differentiation and integration — along with their most important applications. It's designed for people who want to genuinely understand calculus, whether for competitive exams, university courses, or because they're learning machine learning and keep hitting gradient descent without knowing what a derivative actually is.

By the end you'll be able to differentiate any standard function, evaluate definite and indefinite integrals, and understand exactly how calculus powers gradient descent in machine learning. A 30-problem practice set with full solutions is included to reinforce everything covered.

IS THIS RIGHT FOR YOU?

Who Should Attend

✓ THIS IS FOR YOU IF...

- ✓ You were taught calculus mechanically and want to actually understand what derivatives and integrals mean
- ✓ You're studying for JEE, university entrance exams, or a university calculus course and want the conceptual clarity that makes problem-solving easier
- ✓ You're learning machine learning or data science and keep encountering calculus (gradient descent, backpropagation) without understanding the maths
- ✓ You studied calculus years ago and want a fast, rigorous refresher in one session

✗ THIS IS NOT FOR YOU IF...

- ✗ You have no algebra background at all — you need to be comfortable with functions ($f(x) = \dots$) and algebraic manipulation before this session
- ✗ You want advanced calculus (multivariable calculus, vector calculus, differential equations) — this session covers single-variable calculus only
- ✗ You're looking for exam-strategy coaching (specific exam formats, marking schemes) — this is a concept-focused session

Prerequisites: Basic algebra and familiarity with functions (the idea of $y = f(x)$, plotting graphs, solving equations). Trigonometry basics are helpful but not required.

What You'll Cover, Hour by Hour

HOOR 1

Derivatives

- ▶ **What a derivative really means** — the slope idea; instantaneous rate of change visualised geometrically as the limit of a secant line; why this is the central idea of differential calculus
- ▶ **Differentiation rules & shortcuts** — power rule, sum rule, constant multiple rule derived from first principles, then applied fluently across polynomial, exponential, and trigonometric functions
- ▶ **Chain rule, product rule — explained visually** — not just "here's the formula" but a geometric/intuitive explanation of why these rules are necessary and what they're capturing
- ▶ **Applications: rates of change & tangents** — finding the slope of a curve at a point, velocity and acceleration, and the general framework for applying derivatives to real problems

HOOR 2

Integrals

- ▶ **Integration as area accumulation** — the visual definition of a definite integral as the area under a curve; Riemann sums explained; why this is a natural and useful idea
- ▶ **Antiderivatives & the reverse process** — indefinite integration as the operation that reverses differentiation; building intuition for "which function has this derivative?"
- ▶ **Definite integrals & the Fundamental Theorem** — the remarkable connection between derivatives and integrals; why the Fundamental Theorem is the most important result in calculus
- ▶ **Substitution method** — the most important integration technique; how it relates to the chain rule in reverse; worked examples with a range of functions

HOOR 3

Applications & ML Connections

- ▶ **Optimisation problems — maxima & minima** — using derivatives to find where functions reach their maximum and minimum values; the critical point condition and second derivative test
- ▶ **Related rates** — how derivatives let us relate the rates of change of connected quantities; a classic application area for physics and engineering
- ▶ **Where calculus appears in ML (gradient descent)** — the direct line from derivatives to gradient descent; why the derivative of a loss function with respect to a parameter tells you how to update that parameter
- ▶ **Problem-solving session** — working through selected problems from the included problem set, covering the problem-solving strategies that appear in exams and real applications

LEARNING OUTCOMES

What You'll Walk Away With

01 Differentiate any standard function confidently

— polynomials, exponentials, logarithms, trigonometric functions, and their combinations via chain and product rules

02 Evaluate definite and indefinite integrals

— using the Fundamental Theorem and the substitution method for the function types that appear most often

03 Apply calculus to real optimisation problems

— find maxima and minima, interpret rates of change, and use calculus as a problem-solving tool rather than a symbol manipulation exercise

04 Understand why gradient descent works

— the direct connection between taking a derivative and the update rule in every ML optimiser, from SGD to Adam

WHAT'S INCLUDED

Everything You Get

✓ **Recording** — full HD recording of the session shared within 24 hours

✓ **Problem set PDF (30 problems)** — a curated set of problems spanning all topics covered, graded by difficulty from warm-up to exam-level

✓ **Full solutions** — complete worked solutions to every problem in the set, with explanations of the approach at each step

✓ **Learner's Certificate** — digital certificate of completion from Qurious Academy

LOGISTICS

Schedule & Pricing

SCHEDULE

First Saturday of every month

TIME

11:00 AM IST

DURATION

3 hours (single session)

PRICE

₹399

❖ **Is this sufficient preparation for JEE or university calculus exams?**

This session gives you the conceptual foundation and core techniques. For full exam preparation, you'll need to work through the problem set and continue practising. The session builds the understanding that makes practice more effective.

❖ **Do I need to bring pen and paper?**

Yes — calculus is best learned by working through problems as you follow along. Have pen, paper, and ideally graph paper or a graphing calculator ready.

❖ **Is there a maths for ML prerequisite?**

No — this session can be taken standalone. If you're learning ML maths comprehensively, also see the Linear Algebra for ML masterclass and the Maths for ML Power Weekend.

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant has taught mathematics to thousands of students across competitive exam preparation and university-level courses. His approach to calculus is built on a simple conviction: the formulas only make sense once the picture is clear. He's seen too many students who can execute the chain rule without knowing what a derivative is, and this session is his answer to that problem.

His background in both pure mathematics and applied AI engineering gives him a unique perspective: he can teach calculus as a beautiful abstract subject and as the practical engine behind gradient descent — often in the same breath.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

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₹399

Enroll at quriousacademy.com

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Questions? Reach us at hello@quriousacademy.com

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